THE GODEMENT CONSTRUCTION: D AND S SATISFYING THE TWO IDENTITIES MAKE X A SIMPLICIAL ENDOFUNCTOR WITH AUGMENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the checks of (1)(b) and (1)(d) invoke 'the same kind of calculus as in case (1)(a)' (source lines 6505 and 6522) with the resulting simplified expressions stated in-span - the referenced computation is the fully displayed template above them and the outcome is deterministic; the two assumptions of the lemma are used exactly where the simplicial identities demand them ((1)(c) and (1)(e) at j = i).

Editorial note. The label lemma-godement names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The Godement setup: comonad-like data d, s on Y. Same chapter, \label{example-godement}. Lines 6442-6449: the endofunctor Y with natural transformations d : Y -> id and s : id -> Y.
- Characterization of simplicial objects by the five face/degeneracy relations. Same chapter, \label{lemma-characterize-simplicial-object}. Lines 6455-6458: relations (1)(a)-(e) to be verified.
- Interchange of star and circle compositions. Categories chapter, \label{categories-lemma-properties-2-cat-cats}. Lines 6466-6470: (a' circ a) star (b' circ b) = (a' star b') circ (a star b), the engine of every computation.

Source reference. Original lemma 003F: sources/stacks-chow/tags.txt, source lines 132–132. Prerequisite pointer invoked at proof line(s) 6473; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0G5M: sources/stacks-simplicial/pinned-original-source.tex, source lines 6410–6410. Prerequisite pointer invoked at proof line(s) 6444; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 016B: sources/stacks-simplicial/pinned-original-source.tex, source lines 435–435. Prerequisite pointer invoked at proof line(s) 6457; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Example 0G5M if*

$$1_Y = (d \star 1_Y) \circ s = (1_Y \star d) \circ s \quad \text{and} \quad (s \star 1) \circ s = (1 \star s) \circ s$$

then $X = (X_n, d^n_j, s^n_j)$ is a simplicial object in the category of endofunctors of $\mathcal{C}$ and $d : X_0 = Y \to id_{\mathcal{C}}$ defines an augmentation.

Proof. To see that we obtain a simplicial object we have to check that the relations (1)(a) – (e) of Lemma 016B are satisfied. We will use the short hand notation

$$1_a = 1_{X_{a-1}} = 1_Y \star \ldots \star 1_Y \quad (a \text{ factors})$$

for $a \geq 0$. With this notation we have

$$d^n_j = 1_j \star d \star 1_{n-j} \quad \text{and} \quad s^n_j = 1_j \star s \star 1_{n-j}$$

We are repeatedly going to use the rule that for transformations of funtors a, a', b, b' we have $(a' \circ a) \star (b' \circ b) = (a' \star b') \circ (a \star b)$ provided that the $\star$ and $\circ$ compositions in this formula make sense, see Categories, Lemma 003F.

Condition (1)(a) always holds (no conditions needed on d and s). Namely, let $0 \leq i < j \leq n+1$. We have to show that $d^n_i \circ d^{n+1}_j = d^n_{j-1} \circ d^{n+1}_i$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star d \star 1_{n+1-j}) = (1_{j-1} \star d \star 1_{n+1-j}) \circ (1_i \star d \star 1_{n+1-i})$$

We can rewrite the left hand side as

$$\begin{aligned} & (1_i \star d \star 1_{j-i-1} \star 1_{n+1-j}) \circ (1_i \star 1_1 \star 1_{j-i-1} \star d \star 1_{n+1-j}) \\ & = 1_i \star ((d \star 1_{j-i-1}) \circ (1_1 \star 1_{j-i-1} \star d)) \star 1_{n+1-j} \\ & = 1_i \star d \star 1_{j-i-1} \star d \star 1_{n+1-j} \end{aligned}$$

The second equality is true because $d \circ 1_1 = d$ and $1_{j-i} \circ (1_{j-i-1} \star d) = 1_{j-i-1} \star d$. A similar computation gives the same result for the right hand side.

We check condition (1)(b). Let $0 \leq i < j \leq n-1$. We have to show that $d^n_i \circ s^{n-1}_j = s^{n-2}_{j-1} \circ d^{n-1}_i$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j-1} \star s \star 1_{n-1-j}) \circ (1_i \star d \star 1_{n-1-i})$$

By the same kind of calculus as in case (1)(a) both sides simplify to $1_i \star d \star 1_{j-i-1} \star s \star 1_{n-j-1}$.

We check condition (1)(c). Let $0 \leq j \leq n-1$. We have to show $\mathrm{id} = d^n_j \circ s^{n-1}_j = d^n_{j+1} \circ s^{n-1}_j$, i.e.,

$$1_n = (1_j \star d \star 1_{n-j}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j+1} \star d \star 1_{n-j-1}) \circ (1_j \star s \star 1_{n-1-j})$$

This is easily seen to be implied by the first assumption of the lemma.

We check condition (1)(d). Let $0 < j+1 < i \leq n$. We have to show $d^n_i \circ s^{n-1}_j = s^{n-2}_j \circ d^{n-1}_{i-1}$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_j \star s \star 1_{n-2-j}) \circ (1_{i-1} \star d \star 1_{n-i})$$

By the same kind of calculus as in case (1)(a) both sides simplify to $1_j \star s \star 1_{i-j-2} \star d \star 1_{n-i}$.

We check condition (1)(e). Let $0 \leq i \leq j \leq n-1$. We have to show that $s_i^n \circ s_j^{n-1} = s_{j+1}^n \circ s_i^{n-1}$, i.e.,

$$(1_i \star s \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j+1} \star s \star 1_{n-1-j}) \circ (1_i \star s \star 1_{n-1-i})$$

By the same kind of calculus as in case (1)(a) this reduces to

$$(s \star 1_{j-i+1}) \circ (1_{j-i} \star s) = (1_{j-i+1} \star s) \circ (s \star 1_{j-i})$$

If $j = i$ this is exactly one of the two assumptions of the lemma. For $j > i$ left and right hand side both reduce to the equality $s \star 1_{j-i-1} \star s$ by calculations similar to those we did in case (1)(a).

Finally, in order to show that d defines an augmentation we have to show that $d \circ (1_1 \star d) = d \circ (d \star 1_1)$ which is true because both sides are equal to $d \star d$. $\square$